

Teamwork Agreement Template

For the MathWorks Classroom Challenge Projects

Project: Image-Based Defect Detection for Manufacturing Inspection

Team Name: ImageDefect1

Instructor: Kevin Graham

Program/Class: Engineering Pathway Program 2026

Members and Roles

1. Kevin Zuniga - Project Manager & Modeling, Quality Assurance and Reproducibility Lead
2. Flor Serrano - Modeling, Analysis and Validation Lead
3. Jerry Pablo-Ramirez - Modeling, Documentation and Visualization Lead

1) Purpose & Scope

We agree to collaborate to deliver a high-quality, ethical, and reproducible project. Our work will follow the project guidance and produce the specified artifacts (e.g. plots, code, and visualizations).

2) Roles & Responsibilities

- Project Manager (PM): Schedules meetings; maintains timeline; ensures tasks are tracked and disagreements are resolved.
- Modeling Lead: Owns core MATLAB models/simulations
- Analysis/Validation Lead: Performs hand calculations and cross-checks simulation outputs
- Documentation & Visualization Lead: Compiles report (with figures/tables), prepares slides (if needed), and builds visual artifacts (e.g. animations or plots)
- Quality Assurance & Reproducibility Lead: Enforces coherent style across artifacts, ensures work is not overwritten across different versions, ensures data quality, tests/re-runs codes to ensure robustness and reproducibility

3) Communication Norms

- Primary channel: **Discord** (e.g. Slack)
- Secondary channel: **Zoom/email** (e.g. E-mail for summaries/decisions)
- Response time: **24** on weekdays and **48** on weekends (e.g. within 24 hours on weekdays; within 48 hours on weekends)
- Meeting cadence: **1hr** weekly (e.g. weekly working meeting (60–90 minutes on Tuesdays via Zoom))
- Decision records: **Document -> Discord Channel** (e.g. PM logs major decisions in a running document)

4) Collaboration & Tools

- Version control: **Github** (e.g. one shared GitHub repo or Google Drive folder)
- Data management: All datasets, parameters, and assumptions documented in

README.md (e.g. README.md)

- File naming & style: **camelCase with descriptive names**
- Backups: **No backups needed**

5) Quality Standards (Definition of 'Done')

- Requirements meet project description
- At least 75% accuracy for classification
- Learned concepts used in project
- Good documentation

e.g. Requirements meet project description, code is reproducible (runs from the README.md file, all specified figures regenerate, at least one teammate has tested edge cases, all assumptions, units, and references are documented in report)

6) Timeline & Milestones

- Week 1: Problem framing, literature/background review
- Week 2: Dataset/parameters, first pass models (hand calculations + baseline MATLAB scripts); initial plots/visuals.
- Week 3: Validation and quality assurance
- Week 4: Report, slides, presentation rehearsal, repository cleanup

7) Decision-Making & Conflict Resolution

- Default: **PM will hold a dedicated poll/comment section**
- If blocked >48 hrs: **Go with majority vote**
- Technical disagreements: **Evidence-based comparison before vote**
- Escalation: **If unresolved, consult instructor/mentor with a concise memo**

8) Workload, Accountability & Attendance

- Task board: **To-Do**
- Accountability: **Repeated misses trigger role redistribution.**
- Attendance: **If you'll miss a meeting, give 24-hour notice and share updates asynchronously.**

10) Accessibility & Inclusion

We commit to respectful collaboration, equitable turn-taking, and flexibility for documented accessibility needs.

11) Deliverables Checklist

- Inspection function that return pass/fail with a confidence score, evidence overlay and metrics
- Code repository with clear structure and README
- Specific figures & visualizations
- Final report

12) Agreement & Signatures

Name & Signature: **Flor Serrano** Date: **7/9/2026**

Name & Signature: **Kevin Zuniga** Date: **07/9/2026**

Name & Signature: **Jerry Pablo-Ramirez** Date: **7/9/2026**